Functional Annotation Report: MurE (Q88N81, PP_1332) in *Pseudomonas putida* KT2440

Gene/Protein Identity Verification

The research target is confirmed to be correctly identified. UniProt **Q88N81** describes **UDP-N-acetylmuramoyl-L-alanyl-D-glutamate--2,6-diaminopimelate ligase** (EC **6.3.2.13**; gene *murE*; OrderedLocusName **PP_1332**) from *Pseudomonas putida* strain KT2440. The protein belongs to the **MurCDEF family, MurE subfamily**, and carries the diagnostic Mur ligase domains: Mur_ligase_N (IPR000713), Mur_ligase_cen (IPR013221, the central ATP/mononucleotide-binding domain), and Mur_ligase_C (IPR004101). Every piece of the literature consulted matches this identity — the gene symbol *murE*, the EC number, the substrate/product chemistry, the three-domain fold, and the meso-diaminopimelate (m-DAP) substrate are all mutually consistent. There is no gene-symbol ambiguity: *murE* uniformly denotes this ligase across all bacteria.

Note on organism-specific literature: No primary studies of the *P. putida* MurE enzyme itself were found. However, MurE is one of the most highly conserved and best-characterized enzymes in bacteriology. Its function is defined by extensive structural and biochemical work in *E. coli*, *Mycobacterium* spp., and *Corynebacterium glutamicum*, and by demonstrated cross-species functional complementation. Because *P. putida* is a Gram-negative γ-proteobacterium with meso-DAP–type peptidoglycan, the mechanistic conclusions below transfer directly and with high confidence.

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1. Summary (Answer to the Research Question)

MurE (Q88N81 / PP_1332) is a **cytoplasmic, ATP-dependent amide ligase** that catalyzes the **fourth step of the cytoplasmic phase of peptidoglycan biosynthesis**: it adds **meso-diaminopimelic acid (meso-DAP)** as the third residue of the peptidoglycan stem peptide, converting **UDP-MurNAc-L-Ala-γ-D-Glu** (UDP-MurNAc-dipeptide) into **UDP-MurNAc-L-Ala-γ-**

D-Glu-meso-DAP (UDP-MurNAc-tripeptide), with concomitant hydrolysis of ATP to ADP + Pi. It works in the cytoplasm as the third of four sequential Mur ligases (MurC→MurD→**MurE**→MurF), and its product is the obligate substrate for MurF. The reaction is essential for cell-wall assembly and cell viability.

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2. Primary Function: The Catalyzed Reaction and Substrate Specificity

Reaction (EC 6.3.2.13):

> ATP + UDP-MurNAc-L-Ala-γ-D-Glu + meso-2,6-diaminopimelate → ADP + phosphate + UDP-MurNAc-L-Ala-γ-D-Glu-meso-2,6-diaminopimelate

MurE catalyzes "the addition of meso-diaminopimelic acid to nucleotide precursor UDP-N-acetylmuramoyl-l-alanyl-d-glutamate in the biosynthesis of bacterial cell-wall peptidoglycan" (Gordon et al., 2001; P 11124264).

The new bond is an amide linkage between the ε-amino group (the *D*-center amino group) of meso-DAP and the γ-carboxylate of the D-Glu residue already present on the nucleotide precursor.

Substrate specificity. MurE is the amino-acid-adding ligase that selects a **single free amino acid** to place at position 3 of the stem peptide — it is "the last Mur ligase to incorporate a free amino acid" (Rossini et al., 2023; P 36944394).

The identity of that residue is a key taxonomic determinant of peptidoglycan chemistry: - In Gram-negative bacteria (including *Pseudomonas*) and in bacilli/mycobacteria/corynebacteria, MurE is specific for **meso-DAP**. - In many Gram-positive cocci, the orthologous MurE instead adds **L-lysine**.

The structural determinant that dictates this meso-DAP-vs-L-Lys selection was localized in the E. coli MurE crystal structure (Gordon et al., 2001; P 11124264).

As a Gram-negative enzyme with UniProt annotation explicitly naming meso-DAP, *P. putida* MurE is a **meso-DAP-specific** ligase, ensuring the DAP-containing stem peptide required for the direct (A1γ) cross-linking chemistry typical of Pseudomonas peptidoglycan. The 1.4 Å *M. thermoresistibile* MurE·ADP·m-DAP complex directly visualized the m-DAP binding pocket and confirmed UDP-MurNAc-L-Ala-D-Glu (UAG) plus free m-DAP as the physiological substrates (Rossini et al., 2023; P 36944394).

Catalytic mechanism. Like all Mur ligases, MurE uses a two-step, ATP-dependent mechanism: (i) the D-Glu γ-carboxylate is phosphorylated by ATP to form a reactive **acyl-phosphate intermediate** (releasing ADP), and (ii) the meso-DAP amino group performs nucleophilic attack on this intermediate to form the amide bond, releasing inorganic phosphate. Modeling/docking of the substrates supports this acyl-phosphate route, with the substrate carboxylate positioned near the ATP γ-phosphate and the m-DAP amino group oriented for nucleophilic attack (Shanmugam & Natarajan, 2012; P 21491188).
The active site contains a **carbamylated (CO_2-modified) lysine**, as directly observed in the E. coli structure (Gordon et al., 2001; P 11124264),
and catalysis has an **absolute requirement for divalent cations (Mg^{2+})** (Munshi et al., 2013; P 23555903).

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3. Position in the Biochemical Pathway

MurE catalyzes the fourth committed cytoplasmic step of the de novo peptidoglycan (murein) biosynthetic pathway. The Mur ligases act in strict sequence on UDP-N-acetylmuramic acid (UDP-MurNAc):

Step	Enzyme	Residue added	Product
1	MurC	L-Ala	UDP-MurNAc-L-Ala
2	MurD	D-Glu	UDP-MurNAc-L-Ala-γ-D-Glu
3	**MurE**	**meso-DAP**	**UDP-MurNAc-tripeptide**
4	MurF	D-Ala-D-Ala	UDP-MurNAc-pentapeptide

"MurC, MurD, MurE and MurF catalyze the synthesis of the peptidoglycan precursor UDP-N-acetylmuramoyl-L-alanyl-γ-D-glutamyl-meso-diaminopimelyl-D-alanyl-D-alanine by the sequential addition of amino acids onto UDP-N-acetylmuramic acid (UDP-MurNAc)" (Das et al., 2011; P 21445265).

MurE's product (the UDP-MurNAc-tripeptide) is the obligatory substrate of MurF; the resulting UDP-MurNAc-pentapeptide is then transferred by MraY to the membrane carrier undecaprenyl phosphate (forming lipid I), glycosylated by MurG (lipid II), flipped across the inner membrane, and polymerized by penicillin-binding proteins into the mature sacculus. MurE therefore sits at a **non-redundant, committed node** midway through the linear cytoplasmic assembly line. (A parallel recycling route via murein peptide ligase, Mpl, can regenerate UDP-MurNAc-tripeptide from recovered tripeptide, bypassing MurC–E during cell-wall recycling — Das et al., 2011; P 21445265 — but de novo synthesis of the tripeptide is MurE's role.)

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4. Subcellular Localization

MurE functions in the **cytoplasm**. The enzyme is explicitly described as "a cytoplasmic enzyme" (Gordon et al., 2001; P 11124264), consistent with its soluble, nucleotide-sugar substrate and its role in the intracellular (cytoplasmic) phase of peptidoglycan precursor synthesis, which precedes the membrane-associated (lipid I/II) and periplasmic (polymerization/cross-linking) phases. In *Pseudomonas* and other rod-shaped bacteria, the cytoplasmic Mur machinery is functionally coupled to the divisome and elongasome that operate at the inner-membrane/septal interface, but the MurE catalytic reaction itself occurs in the cytosol.

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5. Structure–Function and Genetic Organization

Three-domain architecture. The E. coli MurE structure "consists of three domains, two of which have a topology reminiscent of the equivalent domain found in the … UDP-N-acetylmuramoyl-L-alanine:D-glutamate-ligase (MurD)" (Gordon et al., 2001; P 11124264): an N-terminal domain binding the UDP-MurNAc-peptide, a central Rossmann-like mononucleotide (ATP)-binding domain bearing the P-loop, and a C-terminal domain that contributes to substrate binding and specificity. These three domains map precisely onto the Q88N81 InterPro annotations (Mur_ligase_N, Mur_ligase_cen, Mur_ligase_C), giving strong bioinformatic confirmation of function by homology. Catalysis is accompanied by large **induced-fit domain closure**: binding of UAG and m-DAP triggers "key conformational changes … [and] movements of domains or loop regions" that assemble the productive active site (Rossini et al., 2023; P 36944394).

Genetic context and regulation. *murE* is a member of the conserved **division and cell-wall (dcw / mra) gene cluster**. In mycobacteria a single promoter drives "co-transcription of mur synthetases along with key cell division genes such as ftsQ and ftsW" (Munshi et al., 2013; P 23555903), coupling PG-precursor supply to septation; in *P. putida* KT2440, PP_1332 (*murE*) lies in the analogous *mra/dcw* locus flanked by *murD*, *murF*, *ftsW/ftsI* and *pbp/fts* genes. This organization coordinates MurE expression with the cell-division machinery.

Essentiality. MurE activity is essential: *C. glutamicum murE* was isolated by its ability to complement a temperature-sensitive *E. coli murE* mutant, restoring growth (Wijayarathna et al., 2001; P 11398928) — demonstrating both the conserved enzymatic role and its indispensability for viability.

Because the Mur pathway is bacteria-specific with no human ortholog, MurE is a validated antibacterial drug target (Munshi et al., 2013, P 23555903 ; Hervin et al., 2023, P 38138566 ; Kumar et al., 2023, P 37291783).

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6. Supported and Refuted Hypotheses

Supported: - H1: Q88N81/PP_1332 is a meso-DAP-adding Mur ligase (EC 6.3.2.13) — supported by UniProt annotation, InterPro domains, and cross-species biochemistry/structure. - H2: MurE acts in the cytoplasm as the third amino-acid-adding ligase (MurC→D→E→F) — supported (P 11124264 P 21445265 P 36944394). - H3: The enzyme is ATP- and Mg^{2+}-dependent and uses an acyl-phosphate mechanism — supported (P 23555903 P 21491188 P 11124264). - H4: As a Gram-negative enzyme, it is meso-DAP-specific rather than L-Lys-specific — strongly supported by taxonomy, UniProt naming, and the identified specificity determinant (P 11124264).

Refuted / not supported: - The gene symbol *murE* is **not** ambiguous here; no competing gene of the same symbol was found. The only naming variation is the descriptive substrate (glycyl-D-glutamate in some organisms vs L-alanyl-D-glutamate), which reflects species-specific stem-peptide chemistry, not a different enzyme.

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7. Limitations and Future Directions

- **No direct experimental characterization of the *P. putida* KT2440 MurE** (PP_1332) enzyme (kinetics, structure, knockout phenotype) was located; conclusions rely on strong homology to E. coli/mycobacterial/corynebacterial orthologs and on conserved domain architecture. Direct enzymology (kcat/Km for UAG, m-DAP, ATP) and an experimental *P. putida* structure would confirm the details.
- The precise **specificity residues** governing meso-DAP selection in the *Pseudomonas* enzyme have not been mapped experimentally; they are inferred from the E. coli structural determinant.
- Regulatory coupling (e.g., possible Ser/Thr phosphorylation of Mur ligases seen in Actinobacteria; P 23555903) has not been tested in *Pseudomonas*.

References (PMIDs)

- Gordon E. et al. Crystal structure of UDP-N-acetylmuramoyl-L-alanyl-D-glutamate:meso-diaminopimelate ligase from *Escherichia coli*. P 11124264 .
- Rossini N.O. et al. Crystal structure of *M. thermoresistibile* MurE reveals the binding mode of m-diaminopimelate. P 36944394 .

- Das D. et al. Structure and function of the first full-length murein peptide ligase (Mpl).
  P 21445265 .
- Munshi T. et al. Characterisation of ATP-dependent Mur ligases in *M. tuberculosis*.
  P 23555903 .
- Wijayarathna C.D. et al. Isolation of ftsI and murE genes from *Corynebacterium glutamicum*.
  P 11398928 .
- Shanmugam A. & Natarajan J. Comparative modeling of MurE from *M. leprae* (docking).
  P 21491188 .
- Hervin V. et al. Mur Ligases as an Antibacterial Target (review).
  P 38138566 .
- Kumar S. et al. The Potential of Mur Enzymes as Targets for Antimicrobial Drug Discovery (review).
  P 37291783 .